WAS - Research Project Paper

By: Davit Komyakov

Introduction

On the 28th of October, 1980, United States president Ronald Reagan signed the Commercial Space Launch Act, opening the door to both the acceleration of humanity's progress and knowledge in space. The act was influential enough to spark similar developments in other regions of the world. For example, the European Space Agency (ESA) began working to create reusable rockets with European private companies. Conversely, this piece of legislation has also created monopolies in the space industry and significantly increased negative environmental impacts that accompany space exploration. The debate for whether the commercialization of space exploration and detachment from government authority has contributed to a net positive must be examined through the lens of both countries' government bodies and businesses. With the inevitable formation of independent space organizations such as SpaceX, Blue Origin, etc., the question of ethics and space travel has also been more frequently presented in debates. Furthermore, although it seems to refute a law of this magnitude after its longevity and place in scientific advancement, the government could incorporate policies and measures that would limit the authority of private companies on space travel. This paper will provide a thorough analysis of both the positive attributes of privately owned space exploration and its contributions to science and implications for the future of humanity, but also the potential drawbacks that come along, such as detriments to the environment, creation of monopolies, and problems with ethics from within these organizations.

How has Privatized Space Exploration Worked Well?

Although there are valid critiques and concerns correlated with privatized space exploration, it is difficult to overlook the benefits these companies and private sectors have contributed to science, society, and business as a result of Reagan’s monumental passage of the law. An article from NBC News journalists analyzing current economic impacts of the privatized space industry notes that partnered with advancements in technology quickening scientific development and allowing for easier production and preparation, private companies have significantly decreased launch costs and made private investments into space much more accessible to the general public (Sarlin et al., 2021). This indicates the positive attributes how privatized space exploration has given to the United States by enabling cheaper procedure to conduct scientific research, but also benefit the economy by promoting the emergence of new privately owned businesses in the industry. Furthermore, Increasing the number of active satellites in orbit is significantly beneficial with little downsides as it enables more accessible real-time communication and contributes to scientific progression by not only collecting data of what is beyond Earth’s orbit, but it can scan and provide visuals and data on the environmental state of the Earth (West et al., 2023). Ultimately, satellites help progress science by providing more data on already discovered topics, as well as providing new insights on Earth’s environment and what is beyond. Scientists can apply these discoveries to not only increase public awareness and education on space exploration but also implement scientific discoveries in their argument to garner increased environmental awareness. Therefore, privatized space has further opened the door for economic development and expansion of the industry and its shift as a more vital contributor to the United States economy and allowed for the collection of data to improve the troubled regions of Earth’s environment.

Additionally, privatized space exploration has enabled quicker development and advancement in the space industry and added to humanity's current knowledge of space science than if government organizations were to do it independently. A journal analyzing the benefits of possible future partnerships between private and public space organizations notes that an advantage private organizations hold over the government is that much of the funding for research is not limited to low shares of government funding and taxpayer dollars. Instead, these private companies, such as SpaceX, can acquire funds for missions and research from a wider set of resources, such as private investors, internet services, etc (Rausser et al., 2023). This journal addresses a large negative on an overreliance on government space organizations as funding is largely limited to a very financially demanding sector of research, and instead proposes that private companies can work with private companies to share research and funds for cooperative progress in space exploration. Although this idea would emphasize the weakness in NASA's sources of funding being mainly government funds and an overreliance on private contractors, it would enable the possibility of expanding the diversity in the space funding ecosystem in the United States. An article explaining the benefits of private companies investing in space travel states that over the past 50 years, more than 500 people have been to space, and this number is likely to double in the next 10 years (McCarthy, n.d.). This passage highlights the significant boost of acceleration of space research that has come to the industry of space exploration as a result of expanding government authority over space exploration to private companies. Logically, the prediction is justifiable because not only does decreased prices for space exploration encourage new companies to migrate into the business and contribute to scientific research, but it allows NASA–Still the largest space organization–to partner with the private companies to share inside knowledge and allow for collective growth, rather than limiting private companies to shift into the field with a lack of knowledge and data. Consequently, the article ties in how cheaper launch costs and more companies shifting into the industry promote more people going to space, but what could the implications of this be in terms of commercialized space travel? To analyze the benefits of space tourism, it is crucial to analyze the implications it can have in spreading awareness and education to current and future generations.

In addition, lowered costs of launches and space research, forwarded by the privatization of the space exploration industry, have sponsored the growth of space tourism both now and in the future. Statistics presented by polls surveying Americans of varying generations and gender on their interests in persuring space travel show that although their is a decrease in enterest to tour Earth from an orbital body from older generations, younger demographics between the ages of 18 and 29 are largely open to the idea of touring in space with percentages going all the way up to even (Kennedy and Tyson, 2023). This article shows that although there has been a general decline in interest for space tourism among the general population, the younger generation could be driving customers and create a culture of space tourism that would allow private companies to invest in this sector of space exploration. Consequently, by having the dominant population of the United States adhere to space tourism, it could create a culture and a mindset to be passed down to future generations, which carries on the trend of space tourism. Although there is significant data to support the hypothesis that privatized companies will promote and increase the market for space tourism, it is crucial to understand why this would be a benefit to society as a whole rather than solely the companies involved in the sector.

Space tourism is an overall beneficial extension to space exploration as a whole because of its contributions to research, technological advancements, and increasing the public's awareness of space science. An article explaining the benefits and drawbacks of space tourism highlights how as tourism companies try to remain relevant and keep the business afloat by decreasing the price of flight tickets to customers, they must conduct research and develop more efficient ways to launch and complete cycles with a cheaper price to target a wider demographic (*Everything You Need to Know About Space Tourism*., 2022). Through this conducted research, these private companies will inevitably develop more efficient means of conducting specific sectors of space travel such as the build for the capsule and the launch process. Then, this process becomes evermore beneficial to the entire space industry as the knowledge will eventually be accessed by he public and adopted by other companies that can apply these advancements to space exploration, pushing for scientific research and the betterment of Earth altogether, as explained previously in the paper.

How has Privatized Space Exploration Fallen Short?

Privateized space exploration can also impede the development of scientific progress. A Forbes article captures this idea effectively, stating that if a privately funded company were to have the objective of creating something public, such as the Hubble Space telescope to collect data and progress humanity's knowledge of space, it would need substantial monetary return, which would slow down scientific progress (*The Pros And Cons Of Privatizing Space Exploration*, 2017). This explains how independent organizations conducting research largely separated from government authority can falter as they would require either significant funds to not lose money as a company or to monetize the use of their missions. In the case of a satellite or telescope, if companies such as SpaceX were to attempt to create their rendition of it, they would need to charge for the operator’s service and require payment from other organizations who wish to view images or data, and scientists looking to implement the data into their won theories. Consequently, this would hinder economic growth as funds of this size would be painstakingly difficult to obtain, and the idea of monetizing research equipment, which should otherwise be in the public domain, would not be an asset to private companies in the long term. Therefore, the development of private companies could falter as it would require financial assistance to commit to space research. In a way, this explains a strength that government organizations such as NASA have that private companies do not, and that is all research has been funded through the government and there is no risk of obtaining debt or financial deficit to the organization which receives a set share on a timely basis. Therefore, a large drawback to creating an industry largely run by private companies is that they require significant funds or profit after missions to avoid financial loss and detriment to the company as a whole.

Privatized space exploration's increase in rocket launches and other orbital missions can hurt Earth's environment. A journal of law and social policy published at Northwestern Pritzker School of Law explains that rocket launches are detrimental to the environment as they release black carbon into Earth's atmosphere, contributing to global warming (Nam, 2023). The policy that private companies can operate under a fraction of authority from the government will lead to an acceleration of launches as private companies can collectively lead more missions separately than if the government were to initiate operations independently. It can be argued that although relying on government organizations such as NASA to create contributions to space research through steady and individual missions can slow down humanity's progression in space exploration, it can, without a doubt, be more beneficial to the environment by having a limited number of launches and missions. This summons the debate of whether scientific progression should be hindered to regard for the planet's health and quality. Ultimately, to optimize for ethics and longevity of science, companies should prioritize the condition of the planet over space research as it would be more to the health of the general public and emphasize the idea that scientific research should always cycyle back to the objective of solving critical world problems such as climate change, pollution, and biodiversity loss. This is a reference to the previous part of this study that explains the benefits of having more satellites in orbit. To limit the environmental damages created by quickened progress in space exploration from private companies, stricter national laws are needed. Nations can create international treaties over international waters that all consent to protecting natural resources (Nam, 2023). In doing so, treaties could prevent disagreement in space exploration stemming from the overall objectives of different countries and create transparency in motives. Not only would this limit countries such as the United States and China from harming the environment with excessive missions, but it would also create cooperation between governments that would bleed into benefiting other fields related to natural policy. In conclusion, although space exploration is important to progress, humanities knowledge of space and application of it back to solve objectives on Earth, it is crucial to consider the environmental effects that excessive research can impose on Earth’s population.

Furthermore, privatized space exploration risks the possibility of the formation of monopolies. In a sense, with the access granted to companies to explore space and conduct their operations, too much freedom can overpower limited policies and allow companies to exploit the industry for profit. A BBC article explaining the potential forms of monopolies in the space exploration industry states that dominant corporations could exploit flaws in policies set to create cooperation in the industry and acquire ownership of the Moon and other celestial bodies (Ward, 2019). Although the Outer Space Treaty of 1967 aims to establish the idea that the Moon should be accessible to all identities, there are flaws in the system that do not apply to some organizations and could enable private companies to acquire ownership legally and exploit it. These monopolies discourage the development of the industry by hindering the formation of other private companies to create competition. The lack of competition allows companies to charge any price for data, research, or any benefits of space exploration and limit humanity's access to knowledge by creating a paywall that individuals have no choice but to consent to.

Conclusion

To conclude, the privatization of the space industry, driven by the 1980 Commerial Space Launch Act, has undoubtedly advanced humanity's progression in both technology and the pursuit of new knowledge, as well as real-world applications to settling critical problems ranging from environmental to even international conflicts. To formulate an argument either in support or against the development of a space industry dominated by both government organizations and privately owned companies, one must realize the benefits that both sides present. On one hand, the government can make up for the planning and limit excessive launches while independent companies can work in a sovereign fashion, with the collaboration of some government authority, to accelerate the progression in space exploration and research. Efforts to discourage or hinder either side of the space industry come from an uninformed place, as sufficient research shows the benefits of both and has the potential to patch the problems associated with each if applied together correctly. This idea of collaboration between organizations is highlighted in the Washington Aerospace Scholars readings as well. The reading about the International Space Station (ISS) highlights how much of the space station is comprised of parts from different countries' organizations, such as the United States, Russia, Japan, Germany, etc. As a result, the international space station worked as a bond and a symbol of cooperation between nations and led to collective progress in space research (WAS, n.d.). This explains how collaboration between organizations has the potential to take the positives of what the government does well, as well as the strengths of independence and multitude of companies to optimize humanity's progress and acquisition of information through quickened research in space exploration. A prime example of collaboration of this nature is with the recent Artimis missions driven by NASA partnered with SpaceX and Blue Origin. A Purdue University research paper exploring space policy explains this best by stating that private-public organization collaboration allows the sharing of skillsets and resource use (Guha et al., 2024).

Therefore, privatized space exploration, when regulated with sufficient conditions and operating to advance all of society and eliminate environmental issues, can be a significant asset to advancing humanity's pursuit of knowledge.

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